

Parametrized and Exact Algorithms

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based on course by G. Philip

The main book will be Parametrized Algorithms by Cygan et. al (chapters 1 to 7, 13 and 15). Other than that < copy from moodle >.

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1. Philip's Wisdom

"Why do you want to solve an algorithms problem, probably to make more money. Or kill more people. Or maybe save more people."

2. Brief Intro and Motivation

Parametrized Algorithms starts where the first algorithms course ends: NP Completeness.

Most algorithms we saw in the algorithms course have complexity $O(n^c)$ for some constant c . This raises the question if that is true of all problems, sadly, this is where NP Hardness and Completeness comes in. While we don't know if we can solve them polytime, these NP Complete problems span thousands of fields and experts in their field have tried finding fast algorithms. As these problems are complete, all reduce to each other which implies that if one of them succeed, all of them succeed. Unfortunately, they have all failed, which is the sort of silence that speaks quite loudly.

So how do we cope with this? One is Approx Algorithms. The other idea is Random Algorithms. Another is heuristics and meta-heuristics (taken from physics and biology and what not).

In theory, NP Completeness implies all the problems are equally hard modulo some polynomial time translations. However, some problems were easier to solve in practice.

Example

Type Inference for Hindley-Milner type system is unconditionally not solvable in polynomial time (in the size n of the input programme) as it is hard for the class EXP (that is problems that can be solved in $O(2^{\text{poly}})$ time) (from Maivson, POPL 1990) and $P \subsetneq \text{EXP}$ (Time Hierarchy theorem)

However, we don't think about the size of programme while writing code. And the compiler happily (or sadly) infers the type quite quickly.

Theorem 2.1. *Type inference for Hindley-Milner Type system can be solved in $O(2^k n)$ time where*

- n is the size of input programme,
- k is the nesting depth of types in the programmes

(Lichtenstein and Porveli, POPL 1985)

The idea is to pick something that tends to something that is usually small in practice and limit the exponential blow up to that parameter.

So our roadmap is: pick a parameter k that could be natural or something more convoluted, try to find an algorithm of time complexity $O(f(k)n^c)$. If we succeed the problem is Fixed Parameter Tractable aka FPT.

Definition: Vertex Cover

Vertex cover of a graph G is a subset $S \subseteq V(G)$ such that $G \setminus S$ has no edges.

Definition: Vertex Cover (Standard Parametrization)

Input: Graph G , $k \in \mathbb{N}$

Parameter: k

Question: Does G have a vertex cover of size $\leq k$?

Remark

This can be thought of as choosing a subset as guard posts such that all the roads are guarded.

Also, please make sure to not confuse this with **Dominating Set** (which is choosing vertices to put 1 edge radius umbrellas to cover everyone).

While vertex cover is NP Complete, we can see a simple algorithm that runs in $\mathcal{O}^*(2^k)$ time (we can also do it in $\mathcal{O}^*(1.4656^k)$ time as we'll later see and in $\mathcal{O}^*(1.2529^k)$ as we shall not see).

The idea is that given an edge, one of the two vertices neighboring it have to be in the vertex cover. This is a local property and can be abused to get the algorithm.

VERTEX COVER IN $\mathcal{O}(2^k n)$ TIME

```

1  func existVertexCover( $G, k$ ):
2      |   pick edge  $uv \in E(G)$ 
3      |   return existVertexCover( $G \setminus \{u\}, k - 1$ ) OR existVertexCover( $G \setminus \{v\}, k - 1$ )

```

3. Definitions

Definition: Classical Language

Let Σ be a finite alphabet, for example $\Sigma = \{0, 1\}$ or $\Sigma = \text{ASCII}$. A classical language over Σ is $L \subseteq \Sigma^*$, for example $L = \{\text{All valid Haskell programmes}\}$.

Definition: Parametrized Language

A parametrized language over Σ is $L \subseteq \Sigma^* \times \mathbb{N}$, for example $L = \{(x, k) \mid x \text{ is a valid Haskell programme with } k \text{ parameters}\}$.
For a fixed k , $L_k = \{(x, k) \mid (x, k) \in L\}$ is the k -th slice of L .

Definition: Slice-wise Polynomial Time (XP)

A parametrized language L is slice-wise polynomial time solvable (XP) if there exists an algorithm $\mathcal{A}$ and a computable function f such that:

- For all x, k , $\mathcal{A}$ runs in $\leq |x|^{f(k)}$ time on input (x, k)
- $(x, k) \in L \iff \mathcal{A}((x, k)) = \text{YES}$

Definition: Fixed Parameter Tractable (FPT)

A parametrized language L is fixed parameter tractable (FPT) if there exists an algorithm $\mathcal{A}$ and a computable function f such that:

- For all x, k , $\mathcal{A}$ runs in $\leq f(k) |x|^c$ time on input (x, k) where c is a constant independent of $|x|$ and k .
- $(x, k) \in L \iff \mathcal{A}((n, k)) = \text{YES}$.

For notational convenience, $f(k) |x|^c$ is abbreviated $\mathcal{O}^*(f(k))$.

One motivation could be this table:

$\frac{n^{k+1}}{2^k n}$	$n = 50$	$n = 100$	$n = 150$
$k = 2$	625	2500	5625
$k = 3$	15625	125000	421875
$k = 10$	10^{12}	8×10^{13}	3.7×10^{16}
$k = 20$	1.8×10^{26}	5×10^{30}	2.1×10^{35}

4. Exact Exponential Algorithms

An example could be Hamiltonian Path.

Definition: Hamiltonian Path

Input: Given a graph G and vertex $v \in V$

Question: Does there exist a path P starting at v which covers all vertices of G without repeating vertices or edges.

This problem is NP Hard, however using a DP on sets, we can do this in $\mathcal{O}^*(2^n)$ instead of the brute force $\mathcal{O}^*(n!)$.

Another example is Chromatic Number

Definition: Chromatic Number

Input: Given a graph G and $k \in \mathbb{N}$

Question: Can we properly color the graph G using k colors

While there is a $\mathcal{O}^*(2^k)$ algorithm, we can see an interesting and simple $\mathcal{O}^*(3^k)$ algorithm.

Consider all the subset of vertices and store all the independent sets (that means can be 1 colored). Now consider subsets of vertices and store all the ones that can be partitioned into independent sets (which will take 3^n time as we are basically 3 partitioning the vertices into $S = X \sqcup Y$ and $V \setminus S$) giving us the 2-colorable subsets. Repeat with 3 coloring and so on. This will give an $\underbrace{\mathcal{O}^*(2^n) + \mathcal{O}^*(3^n) + \mathcal{O}^*(3^n) + \dots + \mathcal{O}^*(3^n)}_{k \text{ terms}} = \mathcal{O}^*(3^n)$.

5. Kernalization

TODO. *Introduction with sorting and cycle detection idea*

Example : European Rail Network Problem

Given an input of 1, 40, 000 trains, 25, 000 stations and 16, 00, 000 single train stops (valid trains and station pairs).

Our question is if there is any subset T of trains, they wanted to find a smallest set S of stations such that each train T stops at atleast one of the stations in S .

This is roughly equivalent to the red-blue dominating set which is NP Hard. However, Karsten Werhe gave a set of reduction rules (similar to the cycle detection) that would split this into small cases which he could solve by hand.

TODO. *Section 4.6 of Fundamentals of Parametrized Complexity by Downey and Fellows*

This is a weird place as if we had a polynomial algorithm that could reduce the size of an NP problem by even 1 bit, we could just use the same algorithm to keep reducing it to 1 single bit (which would be the answer).

So what do we do?

TODO. *Parametrization of Kernel*

For vertex cover, we can use the following reduction rules:

- If there is a vertex v of degree zero in G , then delete v from G to get G' . Output: (G', k)
- If there is a vertex v of degree $\geq k + 1$ in G , then delete v from G to get G' . Output: $(G', k - 1)$

Let $(\hat{G}, \hat{k})$ be an instance to which none of the reduction rules applies. We then brute force from there.

We now need to prove:

1. Yes $\iff$ Yes
2. $|\hat{G}| + \hat{k} \leq f(k)$ for some computable function of k .

Proof.

TODO.

■

Definition: Kernelization

Let $L \subseteq \Sigma^* \times \mathbb{N}$ be a parametrized language. A **kernelization** algorithm (also called **reduction to problem kernel**) for L is an algorithm that given an instance (x, k) runs in time polynomial in $(|x| + k)$ and outputs an instance (x', k') such that

1. $|x'| + k' \leq g(k)$ for some computable g
2. $(x, k) \in L \iff (x', k') \in L$

(x', k') is called a kernel of L of size $g(k)$.

First Theorem of Parametrized Complexity 5.1. *A computable parametrized language L has an FPT algorithm if and only if it has a kernalization algorithm.*

Proof. Let there be a $O(f(x))$ (not necessarily polynomial) algorithm for the decision problem where x is the size of problem. ($\Leftarrow$) From the definition of kernalization, we can find the kernel in $O(\text{poly}(|x| + k))$ time and then the solution in time bounded by $O(f(|x| + k)) \leq O(f(g(k)))$ which has no factor of n and hence, is an FPT.

($\Rightarrow$) Suppose $\mathcal{A}$ solves instance (x, k) of L in time $O(f(k) \cdot |x|^c)$.

Algorithm 5.2. Run $\mathcal{A}$ on (x, k) for $|x|^{c+1}$ steps

- If $\mathcal{A}$ stops and returns an answer, return a trivial Yes or No instance
- Else: return (x, k)

It is clear that Yes $\Leftarrow$ Yes. What about the “ $|x'| + k' \leq g(k)$ for some computable g ”?

Well, as $\mathcal{A}$ is an FPT with runtime $O(f(x) \cdot |x|^c)$, the fact the algorithm has not terminated tells us:

$$\begin{aligned} |x|^{c+1} &= O(f(k) |x|^c) \\ \Rightarrow |x|^{c+1} &\leq c_2 f(k) |x|^c \\ \Rightarrow |x| &\leq c_2 f(k) \\ \Rightarrow |x| &= O(f(k)) \end{aligned}$$

Thus, $|x| + k = O(f(k) + k)$.

Thus, we have a kernel of size $O(f(k) + k)$. ■

Notice, the FPT doesn't give a particularly useful kernalization. It is just a nice statement to say.

5.1. Planar Independent Set

Definition: Independent Set

Input: Given a graph G and $k \in \mathbb{N}$

Question: Does there exist $S \subseteq V$ such that $|S| \geq k$ such that $u, v \in S \iff uv \notin E$?

While in the classical setting, this and vertex cover are twins; The same doesn't track here. If we reduce to vertex cover, we have a $\mathcal{O}^*(2^{n-k})$ algorithm which is not FPT.

Sadly, there is no FPT using the standard parametrization as the problem is $W[1]$ hard with respect to k .

However, if we restrict to the planar case, we can however have a kernel.

Definition: Planar Independent Set

Input: Given a planar graph G and $k \in \mathbb{N}$

Question: Does there exist $S \subseteq V$ such that $|S| \geq k$ such that $u, v \in S \iff uv \notin E$?

While this NP Hard (even on bounded degree), we now have a linear kernel (and hence, FPT) on the standard parametrization: if $|V| \geq 4k$ then by 4 color theorem, return **Yes**. Otherwise, return (G, k) .

This is sometimes called a *cheat* kernel as we don't really find out much about the problem.

6. Feedback Vertex Set

Definition: Feedback Vertex Set

Input: Given an undirected graph G on n vertices, $k \in \mathbb{N}$.

Parameter: k

Question: Does G contain a set S of size $\leq k$ such that $G - S$ has no cycles.

We call $S \subseteq V(G)$ is a feedback vertex set of G if and only if $G - S$ is acyclic.

The idea is similar to vertex cover. We first delete all degree 0 and 1 vertices.

Now that we have a graph with degree atleast 2, there must either be a

TODO.

Theorem 6.1. If $n < (2 \log_2 n)^k$ then:

$$(2 \log_2 n)^k < (4k \log_2 n)^k$$

TODO. hain

Proof.

TODO. hain

■

7. Feedback Arc Set

Definition: Feedback Arc Set in Tournament

Input: Given a tournament T on n vertices, $k \in \mathbb{N}$

Parameter: k

Question: Does T contain an arc set F of size $\leq k$, such that $T - F$ has no directed cycle?

Such a F is called a feedback arc set of tournament T .

We could begin with getting rid of all the clear winners (out-degree 0) and clear losers (in-degree 0).

Now the tournament is guaranteed to have a cycle, we use the following claim:

Claim 7.1. An arc (and vertex) in a tournament is part of directed cycle if and only if it is part of a directed triangle.

Proof. FTSOC let the arc uv be part of a the shortest cycle c which is not a triangle.

TODO.

■

Similar to vertex cover, we would like to sort of have a rule to deal with arc's which are part of large number of triangle (k). But we can't delete it as otherwise the instance is no longer a tournament. So what do we do?

Definition

For a subset F of arcs of T , let $T \oplus F$ be the tournament obtained from T by reversing the arcs in F

Claim 7.2. If $T \oplus F$ is acyclic then F is a feedback set of T

Claim 7.3. If F is an inclusion minimal feedback arc set of T then $T \oplus F$ is acyclic.

TODO. *proof is hw*

Corollary 7.4. If (T, k) is an Yes instance of FAST $\iff$ there is a set F of arcs of T , $|F| \leq k$, such that $T \oplus F$ is acyclic.

These gives us clear reduction rules (which are very similar to vertex cover):

- **Reduction Rule 1:** If T has vertex v which is not part of any triangle in T , return $(T - v, k)$
- **Reduction Rule 2:** If T has an arc e that is part of $\geq k + 1$ triangles, then return $(T \oplus \{e\}, k - 1)$

TODO. *Prove Validity of rule 2!*

But why is this a valid kernel?

Proof. If both the reduction rules fail and we are in a non-trivial instance, notice that every arc is part of at most k triangles and every vertex is part of a triangle.

Hence, each arc covers at most k triangles. Thus, reversing an edge can ‘destroy’ at most k triangles which cover $k + 2$ vertices.

Thus, if the instance has $> k(k + 1)$ vertices, we can return a No instance. ■

This gives us a $k^2 + 2k + k = O(k^2)$ kernel.

Idea

The usual idea in coming up with kernelization algorithms is to come up with reduction rules based on ‘simple’ cases.

The way we come up with these rules is to find useful structures we can discover in polynomial time.

Some common structures are:

- Something that is definitely part of the solution
- Something that is definitely not part of optimal solution
- Something which we can replace with a smaller/simpler thing

Note, this same idea gives us a $\mathcal{O}^*(3^k)$ FPT algorithm by branching on the triangles.

$\mathcal{O}^*(2^{O(\sqrt{k} \log(k))})$ is the best known bound for FAST by Alon, Lokshtanov and Saurabh in 2009.

For FVS, the easy bound is $\mathcal{O}^*(3^k)$. The iterated compression method gives $\mathcal{O}^*(2^k)$ and the current best is $\mathcal{O}^*(1.618^k)$ by Kumar¹ and Lokshtanov in 2016.

8. Feedback Vertex Set Revisited

Once we make the reductions and have a graph G left with degree at least 3, consider that if X is a FVS and $G - X$ is a huge forest, then **every leaf in $G - X$ has to have at least 2 edges entering X .**

So we would like at least some high degree vertices in the X . We could formalize this as:

Lemma 8.1. *Let $V(G) = \{v_1, v_2, \dots, v_n\}$ in non-increasing order of degree.*

Let V_{3k} be the first $3k$ vertices in this order (with ties broken arbitrarily). Then, any FVS of G of size $\leq k$, must contain at least one of the vertex of V_{3k}

Proof.

Claim 8.2. Let X be any FVS of G . Then,

¹Another CMI person!

$$\sum_{v \in X} (\deg(v) - 1) \geq m - n + 1$$

Proof. # edges in $G - X \leq n - |X| - 1$

other kind of edges $\leq \sum_{v \in X} \deg(v)$

Adding these, $m \leq \sum_{v \in X} (\deg(v) - 1) + n - 1 \Rightarrow \sum_{v \in X} (\deg(v) - 1) \geq m - n + 1$. ■

FTSOC, let $V_{3k} \subseteq V(G - X)$. Then,

$$\begin{aligned} & \sum_{v \in V_{3k}} (\deg(v) - 1) \\ & \geq 3 \left(\sum_{v \in X} (\deg(v) - 1) \right) \\ & \geq 3(m - n + 1) \end{aligned}$$

Consider

$$\begin{aligned} \sum_{v \notin V_{3k}} (\deg(v) - 1) & \geq \sum_{v \in X} (\deg(v) - 1) \\ & \geq m - n + 1 \\ \sum_{v \in V(G)} (\deg(v) - 1) & \geq 4m - 4n + 4 \\ 2m - n & \geq 4m - 4n + 4 \end{aligned}$$

TODO. *Clarify?!*

Which is a contradiction. ■

This gives us a $\mathcal{O}^*((3k)^k)$ algorithm.

We can also have a randomized $\mathcal{O}^*(4^k)$ algorithm.

TODO. *Photo from Image*

9. Crown Decomposition

Notice, vertex cover was a high degree rule. It worked on high degree vertices. We might also note:

- Degree 0 rule $\rightsquigarrow$ min degree 1 graph: # vertices in the resulting kernel $\leq k + k^2$
- Degree 1 rule $\rightsquigarrow$ min degree 2 graph: # vertices in the resulting kernel $\leq k + \frac{k^2}{2}$
- Degree 2 rule $\rightsquigarrow$ min degree 3 graph: # vertices in the resulting kernel $\leq k + \frac{k^2}{3}$
- $\vdots$

Definition: Matching

A Matching M in Graph G is a set of edges in G of which no two share a common end-point.

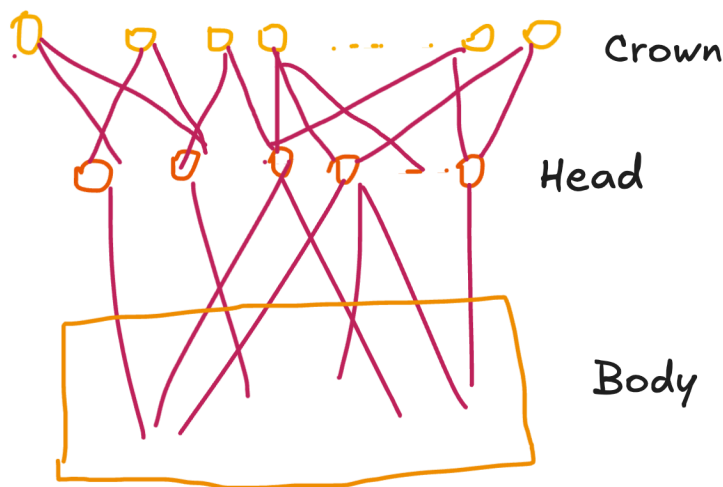
A vertex subset $X \subseteq V(G)$ is **saturated** by matching M if every vertex in X has an edge in M incident to it.

There is a (very, very non-trivial) polynomial time algorithm to find a maximal matching in a graph.

Definition: Crown Decomposition

A **Crown Decomposition** of a graph G is a partition of its vertex set $V(G)$ into three parts: $V(G) = C \uplus H \uplus B$ where:

1. The **crown** C is non-empty independent set in G
2. The set E' of edges between C and the **head** H contains a matching that **saturates** H .
3. There are no edges with one end-point in C and the other in the **body** B .



Claim 9.1. For vertex cover, if $\langle C, H, B \rangle$ is a crown decomposition of a graph G then there is a smallest vertex cover of G that contains **all** of H and **none** of C .

Proof. As there is a matching, **not taking all** of H and taking **none** of C will leave some edge where both its neighbors are not in the vertex cover.

If we don't take all of H and take **some** of C , we can have an equal or better instance by moving the 'guard' from a chosen C to the H it neighbors (can't happen that this doesn't occur as otherwise we have an edge that is not covered). ■

But to do any of that, we need to find the crown-decomposition which can't be polytime (unless $P = NP$). So what do we do?

Konig's Theorem 9.2. *There is a smallest vertex cover of a bipartite graph B , is equal to the size of a maximum matching of B .*

There is a polynomial time algorithm that finds a largest matching, and a smallest vertex cover, on the input bipartite graph B

We now have an(other) algorithm for vertex cover.

A $\mathcal{O}^*(3k)$ KERNEL FOR VERTEX COVER

- 1 Find maximal matching M of G .
 - 2 └ If $|M| > k$ then: return **No**
 - 3 Let V_m be the set of all vertices involved in M . Then $I = (V(G) \setminus V_m)$ is an independent set.
 - 4 └ If $|I| \leq k$ then return (G, k)
 - 5 So: $|I| > k$. Let B be the bipartite graph induced by $V_m \uplus I$. Find a largest matching $\tilde{M}$ and a smallest vertex cover $\tilde{S}$ of B .
 - 6 └ If $|\tilde{M}| > k$ then return **No**
 - 7 Let C be the set of vertices in I that are **not** in $\tilde{S}$, H be the set of vertices of V_m that **are** in $\tilde{S}$ and let $B = V(G) \setminus (C \cup H)$ be the body.
 - 8 └ return $((G \setminus C \cup H), k - |H|)$
-

Our penultimate step indeed choose a crown decomposition as:

1. C is non-empty and independent (as subset of I)
2. H is saturated by a matching from C (otherwise, violates vertex cover property)
3. There are no edges with one end-point in C and the other in **body** B (otherwise, violates vertex cover or induced bipartite).

Remark

There is a linear programme to find a crown decomposition $\langle C, H, B \rangle$ such that $G \setminus C \cup H$ has no crown decomposition.

Furthermore, if $\langle C_1, H_1, B_1 \rangle, \langle C_2, H_2, B_2 \rangle, \dots, \langle C_k, H_k, B_k \rangle$ are the crown decompositions of $G, G \setminus C_1 \cup H_1, \dots, G \setminus (C_1 \cup C_2 \cup \dots \cup C_{k-1}) \cup (B_1 \cup B_2 \cup \dots \cup B_{k-1})$ respectively, then $\langle \bigcup C_i, \bigcup B_i, G \setminus (\bigcup C_i) \cup (\bigcup B_i) \rangle$ is a crown decomposition.

10. An $\mathcal{O}^*(2k)$ Kernel for Vertex Cover

This will be 'optimal' in some sense as Unique Game Conjecture would be violated by, $2 - \varepsilon$ approximation and thus, a kernel better than $(2 - \varepsilon)k$ is not possible.

We will proceed via ILP.

Definition: ILP of Vertex Cover

A variable x_v for each vertex $v \in V(G)$.

For each $uv \in E(G) : x_u + x_v \geq 1$.

For each $v \in V(G)$, $x_v \in \{0, 1\}$

Minimize $\sum_{v \in V(G)} x_v$ subject to above.

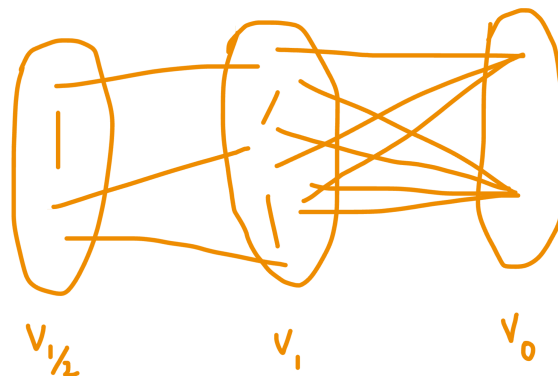
We can relax the integral constraint to get a linear programme

$$0 \leq x_v \leq 1 \text{ for all } v \in V(G)$$

As we shall see in sometime (and blackbox for now), vertex cover's LP has the half-integrability property.

Definition: Half-Integrability

If an ILP admits an optimal solution where every variable takes one of the values $\{0, \frac{1}{2}, 1\}$.



This sort of looks like a crown decomposition already...

Claim 10.1. There is a matching saturating V_1 from V_0

Proof. FTSOC, let there be no such matching. Then, by Hall's marriage lemma, there exists a $X \subseteq V_1$ such that $|N(X)| < |X|$ but then, reassign all variables in $N(X) \cup X$ as $\frac{1}{2}$ and that reduces the objective function.

This doesn't violate constraints as

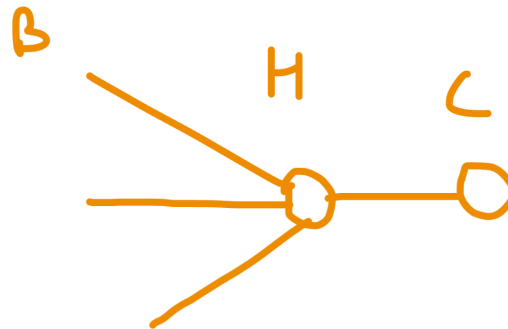
- $X \leftrightarrow N(X)$ as $\frac{1}{2} + \frac{1}{2} = 1$
- $X \leftrightarrow V_1$ as $\frac{1}{2} + \frac{1}{2} = 1$
- $N(X) \leftrightarrow V_1 \setminus X$ as the vertex from $V_1 \setminus X$ was already 1 and $\frac{1}{2} + 1 > 0 + 1 \geq 1$.

Thus, we have a contradiction as the solution was optimal. ■

Notice, $V_{\frac{1}{2}}$ has less than $2k$ vertices as otherwise the relaxed instance has objective greater than the ILP which implies the ILP doesn't admit a solution. Thus, we already have a $\mathcal{O}^*(2k)$ kernel.

11. Sunflower Lemma

Notice, crown decomposition is sort of a generalization of the degree one rule.



Similarly, the sunflower lemma is a generalization of the large degree rule.

Definition: Sunflower

Let $\mathcal{U}$ be a finite universe and let $S_1, S_2, \dots, S_t$ be subsets of $\mathcal{U}$ when:

1. There is a (possibly empty) set C such that $(S_i \cap S_j) = C$ holds for all $1 \leq i < j \leq t$ and
2. $P_i = (S_i \setminus C)$ is non-empty for each $1 \leq i \leq t$.

then $S_1, S_2, \dots, S_t$ is a sunflower with core C and petals $P_1, P_2, \dots, P_t$.

Erdos had asked Rado (who was a kid then, literally eating cereal): Fix positive integers k and d . Does just taking a large enough collection of distinct d -sized subsets of $\mathcal{U}$, guarantee a sunflower with k petals?

Rado answered shortly after leading to the **Sunflower Lemma**.

Theorem (Sunflower Lemma, Erdos & Rado 1960) 11.1. *Let $\mathcal{A}$ be a family of d -sized sets (without duplicates) over a finite universe $\mathcal{U}$. If $|\mathcal{A}| > d!(k-1)^d$ then $\mathcal{A}$ contains a sunflower with k petals. Such a sunflower can be computed from $\mathcal{A}$ in time polynomial $|\mathcal{A}| + |\mathcal{U}| + k$.*